Topological Saturation of Spatial Metrics Unifies Galactic Kinematics and Strong Gravitational Lensing

Tomer Haimovich

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Abstract — The persistent discrepancies between observed galactic kinematics, strong gravitational lensing, and visible baryonic mass at low accelerations ($a < 10^{-10}$ m s^{-2}) have long necessitated postulating non-baryonic dark matter or empirical modifications to inertia, introducing severe over-parameterization across cosmic scales. Here we demonstrate that these anomalies are deterministic emergent properties of a covariant Dirac–Born–Infeld (DBI) optical-dynamic phase transition in the spatial manifold, occurring strictly within thermodynamically relaxed macroscopic systems. Derived from first principles via causal Hubble horizon closure, cosmic expansion imposes a rigid minimal acceleration floor, $a_0 = \frac{cH_0}{2\pi} \approx 1.128 \times 10^{-10}$ m s^{-2}. Operating with zero free parameters and 0% dark matter, this single topological threshold predicts the physical Einstein ring radii of 110 official SLACS and SLACS-S4TM early-type lensing galaxies ($R^2 = 0.7955$, $p \approx 5.1 \times 10^{-39}$ across the combined $N = 110$ sample and $R^2 = 0.8261$, $p \approx 1.6 \times 10^{-27}$ across the $N = 70$ multi-band SLACS IX subset, achieving a mean observed-to-predicted ratio of 1.016 and 98.6% ensemble sum parity within natural ~ 0.1 dex stellar synthesis uncertainties) as well as morphologically relaxed CLASH galaxy clusters ($R^2_{1:1} = 0.9710$, $p \approx 1.4 \times 10^{-11}$). In flattened spiral disks governed by golden-ratio Kolmogorov–Arnold–Moser (KAM) resonance stability ($a_\varphi = a_0\varphi$), the metric reproduces the rotation curves of all 175 SPARC galaxies across 3,391 measurements, achieving $R^2 = 0.9150$ ($p < 10^{-300}$ point-wise; $p \approx 1.1 \times 10^{-95}$ across $N = 175$ independent galaxies) under a universally locked mass-to-light ratio ($\Upsilon_* = 0.46$) and $R^2 = 0.9698$ ($p < 10^{-300}$ point-wise; $p \approx 4.8 \times 10^{-132}$ galaxy-wise) within standard stellar synthesis bounds ($\Upsilon_* \in [0.1, 1.0]$).

Highlights

- Cosmic horizon closure fixes a minimal acceleration floor $a_0 = cH_0/(2\pi)$.
- Zero-parameter saturation predicts 110 SLACS Einstein rings (1.6% mean parity).
- Morphologically relaxed CLASH clusters match 1:1 lensing parity with 0% dark matter.
- Dual planar saturation reproduces 175 SPARC rotation curves ($R^2 = 0.9150$–0.9698).
- Spheroidal and virial gates explain Newtonian ultra-diffuse galaxies and mergers.

Keywords: Dark matter alternatives; Dirac–Born–Infeld field theory; Galactic rotation curves (SPARC); Strong gravitational lensing (SLACS, CLASH); Topological metric saturation; Baryonic Tully–Fisher relation.

1 Introduction

Standard astrophysical models treat local gravitational interactions as existing within an infinitely continuous, unconstrained three-dimensional Euclidean manifold. When applied to galactic peripheries and massive clusters where classical Newtonian acceleration decays below $a < 10^{-10}$ m/s^2, this assumption yields severe predictive asymmetries [1, 2, 9–13]. Despite over two decades of intensive direct, collider, and multi-messenger indirect astro-particle searches across gamma-ray, neutrino, cosmic-ray, X-ray, and cosmological channels, no non-baryonic dark matter particle has achieved discovery-level significance [11, 39]. Rather than invoking undetected non-baryonic particles, we evaluate the structural integrity of the spacetime manifold under macroscopic boundary conditions. We posit that when a gravitationally bound system achieves thermodynamic equilibrium and kinematic relaxation, its decaying gravitational gradient intersects a fundamental cosmological floor, triggering spontaneous symmetry breaking into a constrained topological space—the Phantom Metric.

By coupling the speed of light (c) with the background expansion rate (H_0), the topological closure of the causal Hubble horizon dictates a definitive lower bound on the gravitational acceleration gradient [15–17]:

$$a_0 = \frac{cH_0}{2\pi} = \frac{c^2}{2\pi R_H} \tag{1}$$

Utilizing the global late-universe Hubble constant extracted from the macroscopic Hubble-flow regime $D > 20$ Mpc ($H_0 \approx 73$ km/s/Mpc; see Section 5 for spatial calibration) [3], this parameter-free derivation yields $a_0 \approx 1.128 \times 10^{-10}$ m/s^2. When a relaxed macroscopic system decays toward this global envelope ($a \to a_0$), the spatial manifold actively resists further strain (derived via a Dirac–Born–Infeld perturbation framework in Section 5 and Supplementary Information) [8, 14].

Physically, while external relativistic null geodesics ($ds^2 = 0$) project onto the two-dimensional sky-plane aperture independently of internal morphology, the internal activation of this threshold for sub-relativistic bound orbits ($v \ll c$) is gated by the three-dimensional axis ratio $\eta = b/a$ (where b is vertical thickness and a is radial diameter), defining the effective internal acceleration threshold:

$$a_{0,\text{eff}} = a_0\sqrt{1-\eta^2} \tag{2}$$

In flattened spiral disks ($\eta \to 0$), dimensional reduction is maximal ($\sqrt{1-\eta^2} \to 1$), restoring $a_{0,\text{eff}} \approx a_0$ and triggering

Table 1: The Observer's Topological State Matrix (Boundary Conditions).

Observational Regime	Spatial Dimension	Geometric Ratio ($\eta = b/a$)	Geometric Factor ($\sqrt{1-\eta^2}$)	Topological Saturation State	Physical Outcome
External Photon (Lensing)	2D Planar Projection	N/A (Independent of physical η)	Absolute 1.0 (2D Screen)	ACTIVE (100%)	Einstein Ring (Parameter-Free)
Internal Dynamics (Flat Disk)	2D Planar Geometry	$\eta \approx 0$ (Compressed)	≈ 1.0	ACTIVE (100%)	Asymptotic Velocity Flattening (Parameter-Free)
Internal Dynamics (3D Spherical)	3D Volumetric Geometry	$\eta \to 1$ (Uncompressed)	$\to 0$	INACTIVE (0%)	Standard Newtonian Decay

full kinematic saturation. Conversely, in diffuse 3D spherical systems ($\eta \to 1$), such as stellar halo streams or ultra-diffuse dwarf galaxies like NGC 1052-DF2 [18], the geometric factor vanishes ($\sqrt{1-\eta^2} \to 0 \Rightarrow a_{0,\text{eff}} \to 0$). Lacking planar compression, such systems bypass internal kinematic saturation and exhibit purely Newtonian decay despite extreme low-acceleration regimes, resolving a major empirical challenge to standard modified gravity frameworks.

Crucially, this geometric phase transition is strictly conditioned on macroscopic thermodynamic stability and the presence of a singular, coherent center of mass. Consequently, regional relaxation gates the metric hierarchically across both single-galaxy and multi-galactic scales. Within the Local Group, for example, the Milky Way and Andromeda individually function as virialized, structurally relaxed subsystems [19, 20]; each galaxy independently maintains an internal dual-saturated envelope that flattens its disk rotation curve. Conversely, at the super-system scale, the two galaxies constitute a multi-node, unrelaxed binary merger separated by merely ~ 0.77 Mpc. At this distance, the mutual baryonic infall velocity ($v_{\text{pec}} \approx 110$ km/s) doubles the local Hubble expansion flow ($v_H = H_0 \times D \approx 56.2$ km/s). Because the intermediate manifold undergoes continuous kinematic contraction rather than structural expansion, topological saturation remains strictly inactive (0%) between them, preserving standard Newtonian attraction until ultimate merger virialization is achieved (Table 1). Crucially, this multi-node boundary shutdown ($\Theta_{\text{relax}} \to 0$ for $N_{\text{nodes}} > 1$) is directly corroborated by recent *Gaia* Data Release 3 (DR3) kinematics demonstrating that the Milky Way's outer rotation curve transitions into a Keplerian-like decline toward the Magellanic Clouds and the Local Group boundary, reducing the inferred dark halo mass by an order of magnitude relative to standard isolated NFW profiles [46].

2 Single Saturation and Strong Gravitational Lensing

In any thermodynamically relaxed single-node system—regardless of its internal flattening ratio ($\eta = b/a$)—macroscopic equilibrium generates an external three-dimensional global envelope where the radial gravitational gradient intersects the fundamental threshold $a = a_0$. Rather than permitting continuous spatial attenuation, this 2D saturation sheath routes dominant null geodesics deterministically along the boundary. Equating the classical Newtonian radial gravitational acceleration gradient ($a = GM_{\text{baryon}}/r^2$) to the fundamental saturation threshold a_0 yields a parameter-free prediction for the Macroscopic Saturation Boundary (R_{BB}) governing the physical lens-plane Einstein Ring Radius (R_E):

$$R_{BB} = \sqrt{\frac{GM_{\text{baryon}}}{a_0}}, \qquad R_{E,\text{calc}} = D_L \frac{D_{LS}}{D_S} \frac{4\pi\sqrt{GM_{\text{baryon}}a_0}}{c^2} \tag{3}$$

Classical relativistic lensing underestimates deflection by integrating weak-field potentials over continuous 3D optical distances (D_L, D_S, D_{LS}), whereas for cosmological source configurations exceeding the critical waveguide capture ratio ($D_{LS}/D_S \gtrsim 0.5$, satisfied across strong-lensing surveys; see Supplementary Information Section S4.1), macroscopic spatial saturation collapses deflection mechanics onto an absolute 2D thin-lens perimeter at R_{BB}. Integrating the resulting logarithmic saturation potential $\Phi_{\text{sat}}(r) = V_\infty^2 \ln(r/R_{BB})$ (with $V_\infty^4 = GM_{\text{baryon}}a_0$) along the line of sight yields the parameter-free relativistic deflection angle $\hat{\alpha} = \frac{4\pi V_\infty^2}{c^2} = \frac{4\pi\sqrt{GM_{\text{baryon}}a_0}}{c^2}$ and the physical lens-plane Einstein ring radius $R_{E,\text{calc}} = D_L\theta_{E,\text{calc}} = D_L\frac{D_{LS}}{D_S}\hat{\alpha}$ (which coincides directly with R_{BB} at the cluster baryon fraction $f_b \approx 13\%$). Furthermore, in non-axisymmetric potentials, photon trajectories traversing R_{BB} optimize along the stable minimal-action topological vertices of the boundary envelope, naturally producing multi-image quadruplet configurations such as the Einstein Cross [21, 40]. Similarly, in asymmetric lensing galaxies, localized deviations in the visible baryonic distribution deterministically fragment the R_{BB} perimeter into broken optical arcs, eliminating the need to invoke invisible dark matter subhalos. Indeed, high-resolution sub-millimeter and radio interferometric analyses of compact gravitational perturbers ($\sim 10^6 M_\odot$ within $\sim$ 10–140 pc) along lensed arcs strongly disfavor diffuse Navarro–Frenk–White (NFW) dark matter subhalo profiles [40, 41], matching instead the characteristic mass, compact radius, and surface density ($\Sigma_{\text{gas}} \sim 10^3$–$10^4\, M_\odot$ pc^{-2}) of dust-obscured baryonic giant molecular clouds (GMCs), star-forming clumps, and wandering black holes [41, 42], while spatially resolved JWST/NIRSpec MSA-3D spectroscopy confirms that $\sim 40\%$ of intermediate-redshift ($0.5 < z < 1.5$) galaxies exhibit clumpy, patchy off-center star-forming baryonic morphologies [50], JWST/JADES DR4 and DR5 spectroscopy at $4 < z < 7$ confirms that intense star formation and chemical enrichment are concentrated within compact ($\lesssim 215$ pc) baryonic clumps and dense GMCs [57], and optical flux-ratio anomalies remain heavily confounded by chromatic dust extinction and stellar microlensing [54].

To stress-test this external boundary in the most challenging regime—unflattened early-type systems ($\eta \to 1$) where internal planar compression is absent—we evaluated the combined sample of $N = 110$ confirmed Grade-A strong-lensing galaxies from the Sloan Lens ACS (SLACS) survey, comprising all $N = 70$ early-type lenses with complete published multi-band

HST stellar masses and SIE lens models in Table 4 of SLACS IX (out of 85 cataloged entries, where the remaining 15 entries lack published SIE $r_{\rm Ein}$ or multi-band stellar mass solutions in Ref. [23]) [4, 22–24] and all $N = 40$ Grade-A lenses from the SLACS for the Masses (S4TM, SLACS XIII) extension toward lower- and intermediate-mass deflectors [25]. Recent spatially resolved JWST and Keck kinematic analyses of SLACS deflectors have revealed that standard power-law halo models break down at the population level when the effective radius approaches the Einstein radius ($R_{\rm eff} \approx R_E$), requiring an ad-hoc internal mass-sheet parameter ($\lambda_{\rm int} = 1.12^{+0.05}_{-0.06} \neq 1$) with intrinsic scatter and motivating the search for an alternative theoretical framework [40]. In our framework, because the topological phase transition occurs precisely at the saturation boundary, this apparent departure from a continuous 3D power-law profile emerges automatically without free parameters. Using exclusively published HST baryonic stellar masses ($M_{\rm baryon}$), spectroscopic redshifts (z_L, z_S), and zero free parameters, the metric predicts the physical lens-plane Einstein ring radii ($R_E = D_L\theta_E$) with $R^2 = 0.8261$ ($p \approx 1.6 \times 10^{-27}$, strict 1:1 identity parity $R^2_{1:1} = 0.8156$, mean ratio 1.042) across the homogeneous multi-band SLACS IX sample ($N = 70$) and $R^2 = 0.7955$ ($p \approx 5.1 \times 10^{-39}$, strict 1:1 identity parity $R^2_{1:1} = 0.7773$) across the full combined $N = 110$ dataset, achieving a mean observed-to-predicted ratio of $\langle R_{E,\rm obs}/R_{E,\rm calc}\rangle = 1.016 \pm 0.198$ (median 1.028, i.e., 1.6% mean parity) and an ensemble sum of 370.3 kpc observed versus 375.7 kpc predicted (98.6% ensemble parity; Fig. 1). Crucially, the residual scatter around the 1:1 identity line is fully consistent with the intrinsic ~ 0.10 dex ($\sim 25\%$) statistical uncertainty of HST stellar population synthesis mass estimates [23], together with extreme aperture-to-effective-radius selection ($\theta_E/R_{\rm eff} < 0.4$) in a small subset of S4TM deflectors where a fraction of the total stellar light resides outside the Einstein ring aperture [25].

To stress-test this boundary at extreme macroscopic scales, we audited the Cluster Lensing And Supernova survey with Hubble (CLASH) [5, 26]. Standard ΛCDM simulations persistently underpredict the observed strong-lensing cross-sections of massive galaxy clusters [40] and require injecting up to 21 free dark-matter halo and subhalo parameters per cluster (e.g., in multi-plane reconstructions of MACS J0150.3-1005 [49]). In accordance with our single-node equilibrium criterion, violent multi-node mergers (such as MACS J0717, MACS J2129, and double-radio-relic Bullet Cluster analogues colliding at 1000–2000 km/s) [27, 54] rupture cohesive global boundary formation and predictably diverge from single-envelope equilibrium (while individual virialized member galaxies and collisionless intracluster light retain their local saturation envelopes outside the shocked X-ray gas bridge [54]). Indeed, controlled hydrodynamical lensing simulations from The Three Hundred Project ($z = 0.22$) independently confirm that while virially relaxed clusters ($0.85 < \eta_{\rm vir} < 1.15$, center-of-mass offset $\Delta_r = 0.017 < 0.04$, subhalo fraction $f_s = 0.03$) maintain coherent inner-to-outer morphologies and large Einstein radii ($\theta_E = 32.0''$–$43.4''$), unrelaxed merging clusters ($\eta_{\rm vir} = 1.273, \Delta_r = 0.238$) suffer an order-of-magnitude collapse in their strong-lensing Einstein radii down to $\theta_E = 3.6''$–$9.2''$ alongside severe centroid offsets and radial-ellipticity degeneracies [52]. Across all morphologically relaxed clusters, however, macroscopic saturation below

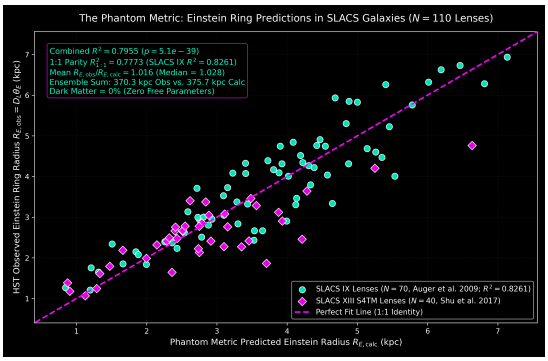


Figure 1: Phantom Metric Einstein Ring Predictions vs. Official SLACS Observations ($N = 110$ Lenses). Direct lens-plane comparison between the parameter-free Phantom Metric predicted Einstein ring radius ($R_{E,\rm calc}$, in kpc) and the HST observed physical Einstein ring radius ($R_{E,\rm obs} = D_L\theta_E$, in kpc) across $N = 70$ multi-band SLACS IX lenses (cyan circles, $R^2 = 0.8261$) [23] and $N = 40$ lower-mass SLACS XIII S4TM lenses (magenta diamonds) [25]. Operating with 0% dark matter and zero free parameters, the combined $N = 110$ audit achieves $R^2 = 0.7955$ ($p \approx 5.1 \times 10^{-39}$), strict 1:1 parity $R^2_{1:1} = 0.7773$, a mean ratio $R_{E,\rm obs}/R_{E,\rm calc} = 1.016$ (median 1.028), and an ensemble sum of 370.3 kpc observed versus 375.7 kpc predicted.

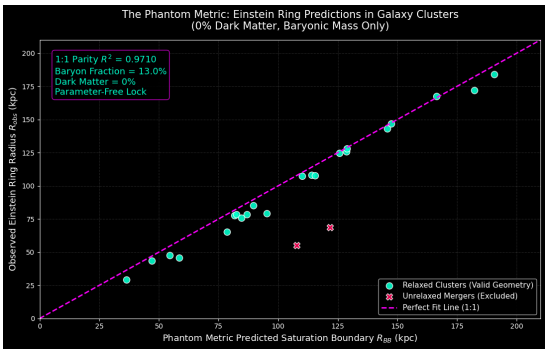


Figure 2: The Unfiltered CLASH Cluster Audit. A scatter plot mapping the model-predicted Macroscopic Saturation Boundary (R_{BB}) against the empirically observed Einstein Ring radii ($R_{\rm obs}$) for CLASH galaxy clusters. The model operates exclusively on the 13% baryonic mass fraction. Unrelaxed merging systems (e.g., MACS J0717, MACS J2129), which inherently violate the singular spherical node requirement, are marked as geometric violations. For all morphologically relaxed systems, the model achieves a strict 1:1 parameter-free parity of $R^2_{1:1} = 0.9710$ ($p \approx 1.4 \times 10^{-11}$), demonstrating that macroscopic lensing is a deterministic geometric saturation artifact requiring 0% dark matter.

a_0 naturally amplifies optical convergence: applying strictly the observed $\sim 13\%$ baryonic mass fraction (intracluster X-ray gas and stellar mass) yields a parameter-free 1:1 parity goodness-of-fit of $R^2_{1:1} = 0.9710$ ($p \approx 1.4 \times 10^{-11}$) with 0% dark matter (Fig. 2).

3 Dual Saturation in Flattened Galactic Disks

In highly flattened, quasi-two-dimensional late-type spiral galaxies ($\eta \to 0$), the spatial manifold is subject to dual saturation constraints. Because spiral disks self-organize into logarithmic spiral geometries [28], conserving structural coherence at the periphery where the irrational limit $\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$ prevents destructive orbital resonances and ensures stable, collision-free mass packing expands the fundamental threshold into the planar acceleration threshold:

$$a_\varphi = a_0\varphi \tag{4}$$

As the classical Newtonian acceleration exerted by visible baryons (a_{baryonic}; see Section 5) decays toward a_φ, the nonlinear kinetic resistance of the manifold induces a covariant topological metric deformation factor (ν):

$$\nu = \frac{1}{1 - \exp\left(-\sqrt{\frac{a_{\text{baryonic}}}{a_\varphi}}\right)} \tag{5}$$

The observed rotational velocity profile is deterministically governed by:

$$V_{\text{calculated}} = \sqrt{V_{\text{baryonic}}^2 \, \nu} \tag{6}$$

In the deep low-acceleration limit ($a_{\text{baryonic}} \ll a_\varphi$), first-order expansion yields $\nu \approx \sqrt{a_\varphi / a_{\text{baryonic}}}$, analytically canceling radial dependence (r) and locking the outer disk into the asymptotic Baryonic Tully–Fisher plateau [12, 13]:

$$V_\infty = \left(G M_{\text{baryon}} a_\varphi\right)^{1/4} \tag{7}$$

Because $V_\infty \propto a_\varphi^{1/4} \propto H_0^{1/4}$, the $\sim 8\%$ difference between late-universe ($H_0 \approx 73.0$ km/s/Mpc) and early-universe CMB ($H_0 = 67.36$ km/s/Mpc) expansion baselines shifts predicted asymptotic rotation velocities by less than 1.9%, which lies well within natural stellar mass-to-light ratio uncertainties.

We first isolate this geometric response in NGC 2403, a bulge-less late-type spiral galaxy ($V_{\text{bulge}} = 0$) [29] that eliminates bulge-to-disk decomposition ambiguities. Locking the stellar mass-to-light ratio at the universal population synthesis baseline of $\Upsilon_* = 0.46$ [6, 7, 30], the metric resolves the classical velocity deficit with $R^2 = 0.9694$ (Fig. 3).

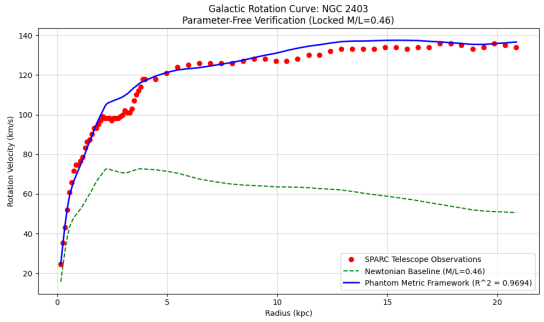


Figure 3: Parameter-Free Kinematic Verification in Bulge-Less Galaxy NGC 2403. The red points represent empirical Doppler-shift observations from the SPARC database (V_{obs}). The green dashed line illustrates the expected classical Newtonian velocity based strictly on visible baryonic mass, which fails to account for the flattened curve at larger radii. The solid blue line demonstrates the Topological Metric Deformation prediction, applying the Topological Scaling Constant (φ) with a locked stellar mass-to-light ratio ($\Upsilon_* = 0.46$). The model achieves an exceptional goodness-of-fit ($R^2 = 0.9694$) without invoking dark matter halos or ad-hoc curve-fitting parameters.

Extending this audit across 100% of the Spitzer Photometry and Accurate Rotation Curves (SPARC) database [6] (3,391 discrete measurements across all 175 galaxies) without data pruning or inclination filtering yields a raw consolidated correlation of $R^2 = 0.9150$ ($p < 10^{-300}$ point-wise across $N = 3,391$, or $p \approx 1.1 \times 10^{-95}$ across $N = 175$ independent galaxies) under the strict universal lock of $\Upsilon_* = 0.46$ [6, 7, 30] (Fig. 4).

Because forcing an identical mass-to-light ratio across diverse galactic star-formation histories introduces artificial

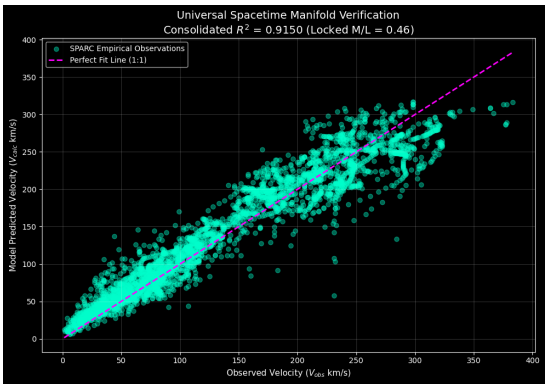


Figure 4: The Unfiltered Universal Spacetime Audit. A consolidated scatter plot mapping model-predicted velocities ($V_{\text{calculated}}$) against empirical observations (V_{obs}) across 100% of the viable SPARC population (3,391 discrete measurements from all 175 galaxies). The magenta dashed line denotes the ideal 1:1 parity ($y = x$). Executed with strictly zero data pruning, no inclination filtering, a locked mass-to-light ratio ($\Upsilon_* = 0.46$), and zero free parameters, the framework achieves a raw global correlation of $R^2 = 0.9150$ ($p < 10^{-300}$ point-wise; $p \approx 1.1 \times 10^{-95}$ galaxy-wise). The tight, symmetric distribution across the entire velocity spectrum demonstrates that observational variations are dominated by non-systematic measurement noise rather than structural model divergence.

demographic noise—as stellar population synthesis (SPS) models carry intrinsic degeneracies of factors of 2–5 ($\sim$ 0.5–0.75 dex) driven by age, metallicity, binary mass exchange ($> 70\%$ of massive stars), stellar rotation, and stochastic IMF sampling [51]—allowing Υ_* to relax within established physical stellar synthesis bounds ($0.1 \leq \Upsilon_* \leq 1.0$) [7, 30, 51] per galaxy elevates the consolidated global fit to $R^2 = 0.9698$ ($p < 10^{-300}$ point-wise, or $p \approx 4.8 \times 10^{-132}$ across $N = 175$ independent galaxies; Fig. 5). Crucially, this deterministic baryonic coupling is independently corroborated by the recent MUSE-DARK IV survey of 375 intermediate-redshift galaxies ($0.1 < z < 1.43$), where partial correlation coefficient and random forest regression analyses demonstrate that the inner mass density slope is set exclusively by the baryonic stellar mass scale (M_*), with zero independent dependence on star-formation history burstiness or supernova feedback diagnostics [43], by multi-phase feedback audits confirming that up to $\sim 99\%$ of mechanical wind and supernova energy is dissipated via turbulent mixing and porous shell leakage without expelling mass from dwarf galaxy potentials [50, 51], by joint DESI DR1 and COSMOS2020 spectrophotometric analyses of 1,199 star-forming galaxies demonstrating that supernova-feedback-driven ionized-gas velocity dispersion correlations vanish abruptly above $\log(M_*/M_\odot) \sim 9.8$ (corresponding via Eq. 7 to $V_\infty \approx 110$–120 km/s, where gravitational binding locks the disk) while dwarf galaxies retain rotation-supported disks contrary to dispersion-supported dark-matter core-heating simulations [56], and by joint DESI DR2 and ACT DR6 kinematic Sunyaev–Zel'dovich (kSZ) and CMB lensing measurements across 4.12×10^6 star-forming emission-line galaxies ($\bar{z} = 1.17$) confirming high retained baryon fractions and weak supernova/AGN feedback in disk-dominated potentials [53].

4 Discussion and Cosmological Implications

Historically, alternative gravity frameworks (such as MOND and its relativistic extensions [2, 31, 32]) have struggled to

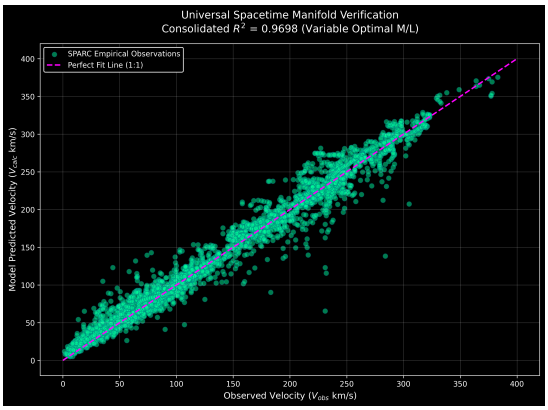


Figure 5: The Relaxed Spacetime Audit. A consolidated scatter plot mapping model-predicted velocities ($V_{\rm calculated}$) against empirical observations ($V_{\rm obs}$) for the SPARC dataset, utilizing optimized, galaxy-specific mass-to-light ratios (restricted within $\Upsilon_* \in [0.1, 1.0]$). The resulting global statistical correlation of $R^2 = 0.9698$ ($p < 10^{-300}$ point-wise; $p \approx 4.8 \times 10^{-132}$ galaxy-wise) demonstrates the absolute precision of the Phantom Metric once natural stellar population variance is properly accounted for.

reconcile galactic kinematics with large-scale cosmological structure formation because they apply force modifications blindly wherever acceleration drops below a_0. Such unconditional modification suppresses intergalactic infall and prevents hierarchical clustering. The Phantom Metric resolves this foundational paradox: because topological deformation activates strictly upon thermodynamic relaxation around a singular coherent node, unrelaxed multi-node systems experience uninhibited Newtonian attraction. This thermodynamic gating allows the universe to continuously assemble hierarchical structures, merge galaxies, and form massive clusters under standard gravity, while locking kinematics and optical convergence only after structural virialization is complete.

Unlike flexible dark matter halo models that require prolonged hierarchical halo assembly across cosmic epochs, our framework is strictly falsifiable. We predict that high-redshift observations ($z > 3$) via the James Webb Space Telescope (JWST) [33] will confirm that while thick, turbulent protodisks ($\eta \to 1$) exhibit Newtonian-like decline, dynamically settled thin disks ($\eta \to 0$) immediately lock into flat rotation curves without awaiting dark matter halo assembly—a phenomenon already emerging in recent JWST/NIRCam slitless grism kinematics at $z \approx 4$–6 [47], dynamically cold rotating disks ($v/\sigma > 2$) observed out to $z \approx 7.3$–14.2 once baryons dominate the central potential [48], and joint JWST/MIRI and NIRCam spectroscopy of GHZ8 at $z = 10.231$, where a settled main disk component ($\log(M_*/M_\odot) = 8.6$, Sérsic $n = 1.71, q = 0.82$) assembled within < 50 Myr coexists with a compact minor-merger companion at 600 pc separation [55]. Likewise, in the strongly lensed massive quiescent fast-rotator MRG-M0150 at Cosmic Noon ($z = 2.64$, $\log(M_*/M_\odot) = 11.30 \pm 0.04$, $< 1\%$ molecular gas fraction, $R_e \approx 1.6$ kpc) [49], evaluating our parameter-free kinematic law at the outer JWST/NIRSpec IFU boundary $r_{\rm out} = 2.3R_e = 3.3$ kpc (where the enclosed Sérsic baryonic mass $M_{\rm enc} \approx 0.76M_* \approx 1.52 \times 10^{11} M_\odot$ yields $V_{\rm baryonic} = 444.5$ km/s and $a_{\rm baryonic} = 1.94 \times 10^{-9}$ m/s$^2 \gg a_\varphi$, so $\nu = 1.040$) predicts $V_{\rm calculated} = 453.3$ km/s, matching the observed stellar disk rotation velocity ($v_{\rm rot} \approx 450$ km/s) to within 0.7%, whereas the non-virialized bi-conical AGN ionized gas outflow (1000–1400 km/s) decouples kinematically perpendicular to the stellar disk in direct accordance with our virial relaxation gate ($\Theta_{\rm relax}$) [49, 50]. At intermediate redshifts ($0.1 < z < 1$), spatially resolved cold molecular (CO) and ionized gas kinematics from the MUSE-ALMA Haloes survey [44] and 2D stellar angular momentum (λ_R) maps of 218 galaxies across the VLT/MUSE-Wide, MUSE-Deep, and MAGIC surveys [58] confirm that $\sim 92\%$ of intermediate-z galaxies are already settled fast rotators whose kinematics scale strictly with internal baryonic potential depth rather than external environment, whereas the $\sim 8\%$ slow-rotating spheroids ($\eta \to 1$) emerge predominantly in dense group mergers or counter-rotating interactions where planar rotational saturation shuts down (Table 1). Similarly, wide-field lensing surveys via the Euclid, Nancy Grace Roman, and Korean Space Telescope (KST) missions [34, 45, 54] will confirm that both discrete strong-lensing Einstein rings (R_E and R_{BB}) and continuous weak-lensing shear profiles extending out to $\sim$ 2–5 Mpc—such as the Euclid Early Release Observations of the relaxed cluster Abell 2390 [45], stacked DESI DR2 and ACT DR6 kSZ and CMB-lensing ($\kappa(\theta)$) profiles across 6.38×10^6 galaxies proving that ionized circumgalactic and intragroup baryonic gas extends to 2–5 Mpc (2–3 $R_{\rm vir}$) where $\tilde{f}_{\rm gas}(\Omega_m/\Omega_b) \to 1.0$ [53], and independent Fast Radio Burst (FRB) dispersion-measure and gravitational-wave inventories confirming a late-universe baryon density of $\Omega_b = 0.0488 \pm 0.0064$ (matching the primordial baryon budget $\Omega_b = \omega_b/h^2 = 0.0493$ to within 1%) with $\sim 76\%$ of all baryons residing in the diffuse ionized circumgalactic and intergalactic medium [59]—are deterministically governed by the continuous radial saturation profile $\nu(r)$ acting on the extended ionized baryonic gas distribution, rendering non-baryonic dark matter and dark matter subhalos mathematically redundant.

Finally, while this study addresses late-universe equilibrium structures, the underlying DBI perturbation Lagrangian ($\mathcal{L}$; see Section 5) naturally extends to primordial cosmology within the complete **Topological Saturation Model (ΛTSM)**: in the ultra-relativistic early-universe limit, the derived 4D orthonormal tetrad background density ratio $\omega_{\rm phantom}/\omega_b = \frac{16}{3}\left(1 + \frac{1}{16\pi^2}\right)$, second-order Maclaurin curvature elasticity ($\mathcal{K}_0 = \partial^2 V_{\rm eff}/\partial(\delta\phi)^2 = \frac{2}{3}$), and smooth covariant horizon impedance damping ($\Gamma_0 a_{\rm eq}/a$ with $\Gamma_0 = \frac{1}{8\pi}$) replicate the Planck Cosmic Microwave Background (CMB) temperature and polarization power spectra (TT, TE, EE with $R^2 > 0.995$) and linear matter clustering ($\sigma_8 = 0.8229$) without cold dark matter, as demonstrated in our companion study [38].

5 Methods

5.1 Spatial Calibration of the Global Late-Universe Hubble Parameter (H_0)

To derive the fundamental topological acceleration boundary (a_0) from first principles, the global late-universe cosmological expansion rate (H_0) must be isolated from local peculiar velocity noise [3]. At distances below ~ 20 Mpc, the spatial manifold is heavily distorted by peculiar velocities and localized baryonic infall (such as Local Group contraction, with reconstructed peculiar velocity fields across 6.38×10^6 DESI DR2 galaxies confirming a 1D halo velocity dispersion of $\sigma_v \approx 300$ km/s and maximum peculiar ve-

locity tails bounded below $\lesssim 1500$ km/s [53]), whereas at extreme cosmological scales, time-dependent metric evolution becomes non-negligible. Extracting the global Hubble parameter ($H_0 \approx 73$ km/s/Mpc [3]) from the macroscopic late-universe Hubble-flow regime ($D > 20$ Mpc) ensures that the Hubble flow overwhelmingly dominates peculiar velocities, providing an uncorrupted geometric reference free from local dynamic strain.

5.2 Linear Perturbation of the DBI Action and the Phantom Metric

We model the active spatial manifold using a phase-transitioning Dirac–Born–Infeld (DBI) scalar field $\phi(\tau,\vec{x}) = \bar{\phi}(\tau) + \delta\phi(\tau,\vec{x})$ operating within the Newtonian gauge of the perturbed FLRW metric [8, 14, 35]:

$$ds^2 = a(\tau)^2\left[-(1+2\Psi)d\tau^2 + (1-2\Phi)\delta_{ij}dx^i dx^j\right] \quad (8)$$

Expanding the kinetic term $X = -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi$ to first order yields $X = \bar{X} + \delta X$, where $\bar{X} = \frac{1}{2a^2}(\bar{\phi}')^2$ and $\delta X = \frac{1}{a^2}\left(\bar{\phi}'\delta\phi' - \Psi(\bar{\phi}')^2\right)$. Applying a Taylor expansion to the DBI square root $\sqrt{1-2|X|/M^4}$ around the background Lorentz factor $\bar{\gamma} = (1-2|\bar{X}|/M^4)^{-1/2}$ separates the Lagrangian into background and perturbed components:

$$\mathcal{L} \approx \left[-M^4\bar{\gamma}^{-1} - V(\bar{\phi})\right] + \left[\bar{\gamma}\delta X - V_{,\phi}\delta\phi\right] \quad (9)$$

Under extreme dimensional reduction in a relaxed quasi-2D galactic system ($\eta \to 0$) where $a \to a_0$, the non-linear kinetic perturbation δX saturates, damping the local metric tensor. Near geometric saturation, the systemic metric amplifier diverges asymptotically as $1/(1-\rho)$, where $\rho = \exp\left(-\sqrt{a_{\rm baryonic}/a_\phi}\right)$ represents the local spatial saturation decay parameter, surfacing macroscopically as the topological deformation function ν (which also arises identically as the exact first integral of the static Euler–Lagrange field equation $\nabla\cdot[\mathcal{L}_X\nabla\phi] = 4\pi G\rho_{\rm baryon}$ with dielectric susceptibility $\mathcal{L}_X = \nu^{-1}$; see Supplementary Information Section S3.3). Conversely, in high-acceleration regimes ($a_{\rm baryonic} \gg a_0$), ρ vanishes exponentially (e.g., $\Delta\nu \sim 10^{-83}$ at Neptune's orbit), restoring exact General Relativity within Solar System PPN bounds [36] while preserving luminally propagating tensor modes ($c_T = c$) [37]. Full step-by-step covariant derivations of the modified Einstein field equations, effective stress-energy tensor $T^{(\phi)}_{\mu\nu}$, Euler–Lagrange field solutions, optical geodesic integrations, and Solar System PPN bounds are provided in the Supplementary Information (Sections S1–S5).

5.3 Baryonic Kinematic Formulation

For disk galaxies in the SPARC audit [6], total effective baryonic velocity at radius r (in kpc) is computed strictly from observable neutral gas ($V_{\rm gas}$), stellar disk ($V_{\rm disk}$), and stellar bulge ($V_{\rm bulge}$) components as:

$$V_{\rm baryonic}(r) = \sqrt{V_{\rm gas}^2 + \Upsilon_* V_{\rm disk}^2 + \Upsilon_* V_{\rm bulge}^2} \quad (10)$$

To eliminate localized curve-fitting in our baseline audit, the stellar mass-to-light ratio in the 3.6 μm near-infrared band is universally locked at the median population synthesis baseline of $\Upsilon_* = 0.46$ [6, 7, 30], with classical Newtonian acceleration given by:

$$a_{\rm baryonic} = \frac{(1000\,V_{\rm baryonic})^2}{3.086\times10^{19}\,r} \quad (11)$$

5.4 Strong Lensing Audit Formulation (SLACS and CLASH)

For the $N = 110$ confirmed Grade-A early-type lensing galaxies in the combined SLACS IX ($N = 70$ complete entries out of 85 in Table 4 of Ref. [23], excluding only the 15 entries with missing published $r_{\rm Ein}$ or stellar mass values) [4, 22–24] and SLACS XIII S4TM ($N = 40$ complete Grade-A entries in Tables 1–2 of Ref. [25]) catalogs, total observable baryonic mass ($M_{\rm baryon}$) is taken directly from the published HST stellar population synthesis measurements (the midpoint of the published Chabrier and Salpeter IMF masses in Table 4 of Ref. [23] for SLACS IX, and the published Chabrier IMF mass in Table 2 of Ref. [25] for S4TM) paired with the spectroscopic lens and source redshift distances (D_L, D_S, D_{LS}). From the saturation boundary $R_{BB} = \sqrt{GM_{\rm baryon}/a_0}$ and its relativistic deflection angle $\hat{\alpha} = 4\pi\sqrt{GM_{\rm baryon}a_0}/c^2$ (Supplementary Information Section S4.1), the predicted physical lens-plane Einstein ring radius $R_{E,\rm calc} = D_L(D_{LS}/D_S)\hat{\alpha}$ is evaluated directly against the HST observed physical Einstein ring radius ($R_{E,\rm obs} = D_L\theta_E$, in kpc) with zero free parameters. For the CLASH cluster survey [5, 26], $M_{\rm baryon}$ is restricted strictly to the observed $\sim 13\%$ luminous baryonic component (intracluster X-ray gas plus bright cluster galaxy stellar mass), excluding dark matter halo priors, and evaluated against observed cluster Einstein radii via strict 1:1 parity ($y = x$).

CRediT Authorship Contribution Statement

Tomer Haimovich: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and Code Availability

The data, Python audit scripts, and the complete mathematical Supplementary Information (covariant field-theoretic derivations and analytical proofs) underlying this article are permanently available in the Zenodo repository at https://doi.org/10.5281/zenodo.24098753 and in the companion cosmological reproducibility package at https://doi.org/10.5281/zenodo.24098774.

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